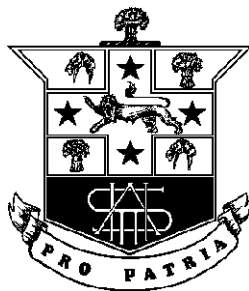


HURLSTONE AGRICULTURAL HIGH SCHOOL



YEAR 12

MATHEMATICS EXTENSION 2

2008

HSC COURSE

HALF YEARLY EXAMINATION (ASSESSMENT TASK 2)

EXAMINERS ~ G. RAWSON AND J. DILLON

GENERAL INSTRUCTIONS

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| <ul style="list-style-type: none">• Reading Time – 5 minutes.• Working Time – Two Hours.• Attempt all questions.• Questions are of equal value.• All necessary working should be shown in every question.• This paper contains Five (5) questions.• Total Marks – 75 marks | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved Calculators and Templates may be used.• Each question is to be started in a new Examination Booklet.• This assessment task must NOT be removed from the Examination Room. |
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STUDENT NAME / NUMBER:

TEACHER:

QUESTION 1: (USE A SEPARATE ANSWER BOOKLET)

MARKS

- (a) Two complex numbers are given by
 $z = 3 - 4i$ and $w = 2 - 2i$
- (i) Find, in the form $x + iy$, the product $\bar{z}w$. **1**
- (ii) Find, in the form $x + iy$, the two square roots of z . **2**
- (iii) (α) Express w in modulus – argument form. **1**
- (β) Find, in the form $x + iy$, w^5 . **2**
- (b) The locus of a point P which moves in the complex plane is represented by the equation
$$|z - (3 + 4i)| = 5$$
- (i) Sketch the locus of the point P **1**
- (ii) (α) Find the maximum value of the modulus of z . **1**
- (β) Write down the value of $\arg z$ when P is in the position of maximum modulus. **1**
- (γ) Find the value of the modulus of z when $\arg z = \tan^{-1}\left(\frac{1}{2}\right)$ **1**
- (c) The complex number z is given by $z = 1 - i$. **2**
Find real numbers a and b such that $az = 1 - \frac{b}{z}$.
- (d) In an Argand diagram the point P represents the complex number z and the point Q represents the complex number w and the point R represents the complex number $z - w$ with O being the origin.
It is given $z - w = iz$.
- (i) Describe the geometric properties of triangle POR , providing full reasoning for your answer. **2**
- (ii) Find the size of angle QPR . **1**

QUESTION 2: (USE A SEPARATE ANSWER BOOKLET)

MARKS

Consider the function $g(x) = 4\sqrt{x} - 2x$.

- | | | |
|-------|--|----------|
| (a) | Write down the domain of $g(x)$. | 1 |
| (b) | Find the x intercepts of the graph of $y = g(x)$. | 1 |
| (c) | Show that the curve $y = g(x)$ is concave downwards for all $x > 0$. | 1 |
| (d) | Find the coordinates of the stationary point and determine its nature. | 1 |
| (e) | Sketch the graph of $y = g(x)$, clearly showing all essential features. | 1 |
| (f) | Hence, by consideration of your graph of $y = g(x)$, sketch each of the following on separate diagrams, showing all essential features. | |
| (i) | $y = g(x) $ | 2 |
| (ii) | $y = g(x - 2)$ | 2 |
| (iii) | $y = g(x)$ | 2 |
| (iv) | $ y = g(x)$ | 2 |
| (v) | $y = \frac{1}{g(x)}$ | 2 |

QUESTION 3: (USE A SEPARATE ANSWER BOOKLET)

MARKS

- (a) A polynomial has a remainder of 5 when divided by $(x-3)$ and a remainder of 12 when divided by $(x-4)$. 2
What is the remainder when the polynomial is divided by $(x-3)(x-4)$?
- (b) The polynomial $P(x) = x^3 + 3x - 5$ has roots α , β and γ . 2
Find the value of $\alpha^3 + \beta^3 + \gamma^3$.
- (c) The equation $x^3 + 3x^2 - 2 = 0$ has roots α , β and γ . 3
Find the cubic equation with roots α^2 , β^2 and γ^2 .
- (d) Given that $x = 2 + i$ is a root of the polynomial $P(x) = x^3 - 2x^2 - 3x + 10$, 3
factorise $P(x)$ over the complex field, $\mathbb{C}$.
- (e) If $P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$ has a triple zero, find all the zeros of $P(x)$. 2
- (f) Consider the function $P(x) = x^3 + ax + b$. 3
Show that the function will have three distinct roots if $4a^3 + 27b^2 < 0$.

QUESTION 4: (USE A SEPARATE ANSWER BOOKLET)

MARKS

- (a) The ellipse $\mathcal{E}$ has the equation

$$4x^2 + 9y^2 = 36$$

- (i) Write down:

(α) its eccentricity **1**

(β) the coordinates of its foci S and S' **1**

(γ) the equation of each directrix **1**

(δ) the length of the major axis. **1**

- (ii) Sketch the ellipse $\mathcal{E}$. Show the x and y intercepts as well as the features found in parts (β) and (γ) of part (i) above. **1**

- (iii) Without using the formula for the area of an ellipse, show by integration that the area of the ellipse $\mathcal{E}$ is 6π square units. **4**

- (b) (i) Show that if $y = px + q$ is a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ then } p^2 a^2 - b^2 = q^2 \quad \text{3}$$

- (ii) Hence find the equations of the tangents from the point $(1, 3)$ **3**
to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$.

QUESTION 5: (USE A SEPARATE ANSWER BOOKLET)

MARKS

- (a) (i) Use De Moivre's theorem, or otherwise, to show that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$. 2

- (ii) Hence, solve $8x^3 - 6x - 1 = 0$. 2

- (iii) Deduce that $\sin \frac{\pi}{18} = \sin \frac{7\pi}{18} + \sin \frac{11\pi}{18}$. 1

- (b) The polynomial $P(x) = x^3 + qx^2 + rx + s$ has real coefficients. It has three distinct zeros α , $-\alpha$ and β .

- (i) Prove that $qr = s$. 3

- (ii) The polynomial does not have three real zeros. Show that two of the zeros are purely imaginary. (A number is purely imaginary if it is of the form iy , with y real and $y \neq 0$.) 2

- (c) You are given

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A\end{aligned}$$

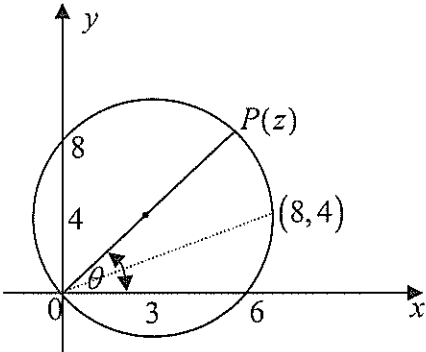
and

$$\sin 2A = 2\sin A \cos A$$

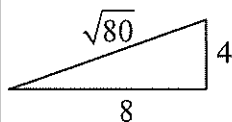
If n is any integer, prove that:

- (i) $\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \cos^n \frac{\pi}{n}$. 2

- (ii) $(1 + i\sqrt{3})^{2n} + (1 - i\sqrt{3})^{2n}$ has either the value 2^{2n+1} or -2^{2n} , according as to whether n is or is not a multiple of 3. 3

Year 12	Mathematics Extension 2	Half Yearly Examination 2008
Question 1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	uses the relationship between algebraic and geometric representations of complex numbers and of conic sections	
Part	Solutions	Marking Guidelines
(a) (i)	$\bar{z}w = (3+4i)(2-2i)$ $= 14 + 2i$	Award 1 correct solution
(ii)	Let $\sqrt{3-4i} = a+ib$ $\therefore 3-4i = a^2 - b^2 + 2abi$ $\therefore a^2 - b^2 = 3 \quad \dots\dots (1)$ $2ab = -4 \Rightarrow ab = -2 \quad \dots\dots (2)$ Since $(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$ $= 9 + 16$ $= 25$ Then $a^2 + b^2 = 5 \quad \dots\dots (3)$ $(1) + (3) \Rightarrow 2a^2 = 8$ $\therefore a = \pm 2, b = \mp 1$ The two square roots are $2-i, -2+i$.	Award 2 correct solution Award 1 establish initial equations only
(iii)(α)	$2-2i = 2\sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right) = 2\sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right) \right)$	Award 1 correct solution
(β)	$w^5 = \left(2\sqrt{2} \right)^5 \left(\cos\left(-\frac{5\pi}{4}\right) + i\sin\left(-\frac{5\pi}{4}\right) \right)$ $= 128\sqrt{2} \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$ $= -128 + 128i$	Award 2 correct solution Award 1 attempts to apply De Moivre's Theorem
(b) (i)		Award 1 correct sketch
(ii) (α)	Maximum value of $ z $ is 10 (i.e. a diameter)	Award 1 correct solution
(β)	When P is in maximum modulus $\arg z = \theta = \tan^{-1} \frac{8}{6} = \tan^{-1} \frac{4}{3}$	Award 1 correct solution

(γ)



When $\arg z = \tan^{-1} \frac{1}{2}$, $|z| = \sqrt{80} = 4\sqrt{5}$

Award 1

correct solution

(c)

$$az = 1 - \frac{b}{z}$$

$$\begin{aligned} \therefore a(1-i) &= 1 - \frac{b}{1-i} \\ &= 1 - \frac{b(1+i)}{2} \\ &= \frac{2-b-bi}{2} \end{aligned}$$

$$\therefore a - ai = 1 - \frac{b}{2} - \frac{b}{2}i$$

$$\therefore a = 1 - \frac{b}{2} \text{ and } a = \frac{b}{2}.$$

Solving simultaneously gives

$$a = \frac{1}{2}, b = 1.$$

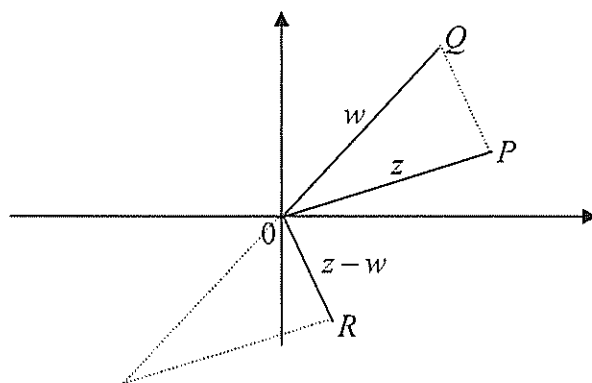
Award 2

correct solution

Award 1

substitution and attempt to realise the denominator

(d) (i)



If $z - w = iz$, then we need to rotate P about O through $\frac{\pi}{2}$.

$$\therefore \angle POR = \frac{\pi}{2}$$

$$|z - w| = |iz| = |z|$$

$$\therefore OR = OP$$

$\therefore \triangle OPR$ is right-angled isosceles.

Award 2

correct solution ~ with full reasoning

Award 1

correct approach ~ with incorrect of insufficient reasoning

(ii)

$$\angle OPR = \frac{\pi}{4} \text{ and } \angle OPQ = \frac{\pi}{2}$$

$$\therefore \angle QPR = \frac{3\pi}{4}$$

Award 1

correct solution

Year 12	Mathematics Extension 2	Half Yearly Examination 2008
Question 3	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials	
Part	Solutions	Marking Guidelines
(a)	Let $P(x) = (x-3)(x-4)Q(x) + ax + b$ $P(3) = 5 = 3a + b \dots (1)$ $P(4) = 12 = 4a + b \dots (2)$ $(2) - (1) \quad a = 7 \text{ and } b = -16.$ $\therefore \text{Remainder} = 7x - 16.$	Award 2 correct solution Award 1 uses remainder theorem
(b)	Let $x = \alpha, \alpha^3 + 3\alpha - 5 = 0$ Let $x = \beta, \beta^3 + 3\beta - 5 = 0$ Let $x = \gamma, \gamma^3 + 3\gamma - 5 = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha + \beta + \gamma) - 15 = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 + 3.0 - 15 = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 = 15.$	Award 2 correct solution Award 1 substantial progress towards solution
(c)	Let $y = x^2 \Rightarrow x = \sqrt{y}$ The desired equation is given by $(\sqrt{y})^3 + 3(\sqrt{y})^2 - 2 = 0$ $y\sqrt{y} + 3y - 2 = 0$ $y\sqrt{y} = 2 - 3y$ $y^3 = 4 - 12y + 9y^2$ $\therefore y^3 - 9y^2 + 12y - 4 = 0$	Award 3 correct solution Award 2 substantial progress towards solution Award 1 establishes required transformation
(d)	If $x = 2 + i$ is a root, then $x = 2 - i$ is also a root. $\therefore (x - (2 + i))(x - (2 - i))$ $= ((x - 2) - i)((x - 2) + i)$ $= x^2 - 4x + 5$ is a factor. $\therefore x^3 - 2x^2 - 3x + 10 = (x^2 - 4x + 5)(x + 2)$ by inspection. $\therefore x^3 - 2x^2 - 3x + 10 = (x - 2 + i)(x - 2 - i)(x + 2)$ over $\square$.	Award 3 correct solution Award 2 determines quadratic factor Award 1 uses conjugate root theorem
(e)	$P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$ $P'(x) = 4x^3 + 6x^2 - 24x + 14$ $P''(x) = 12x^2 + 12x - 24 = 12(x^2 + x - 2) = 12(x + 2)(x - 1)$	Award 2 correct solution Award 1 uses multiple root theorem ~ does not determine other root

(f)	By observation $P''(1) = P'(1) = P(1) = 0$ $\therefore x = 1$ is the triple root. $\therefore 1 + 1 + 1 + \alpha = -2$ $\therefore \alpha = -5$. $\therefore$ Roots are 1, 1, -5. $P(x) = x^3 + ax + b$ $P'(x) = 3x^2 + a$ Stationary points occur @ $P'(x) = 0$ $\therefore x = \pm \sqrt{-\frac{a}{3}}$ $P\left(\sqrt{-\frac{a}{3}}\right) = b + \frac{2}{3}a\sqrt{-\frac{a}{3}} \dots\dots (y_1)$ $P\left(-\sqrt{-\frac{a}{3}}\right) = b - \frac{2}{3}a\sqrt{-\frac{a}{3}} \dots\dots (y_2)$ 3 distinct roots $\Rightarrow y_1 y_2 < 0$ $\left(b + \frac{2}{3}a\sqrt{-\frac{a}{3}}\right)\left(b - \frac{2}{3}a\sqrt{-\frac{a}{3}}\right) < 0$ $b^2 - \frac{4}{9}a^2 \cdot -\frac{a}{3} < 0$ $b^2 + \frac{4a^3}{27} < 0$ $27b^2 + 4a^3 < 0$	Award 3 correct solution Award 2 substantial progress towards solution Award 1 establishes the x coordinates of the SP only
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Year 12	Mathematics Extension 2	Half Yearly Examination 2008
Question 5	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E2	chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings	
E3	uses the relationship between algebraic and geometric representations of complex numbers and of conic sections	
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials	
Part	Solutions	Marking Guidelines
(a) (i) E2 E3	$(\cos \theta + i \sin \theta)^3$ $= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3i^2 \cos \theta \sin^2 \theta + i^3 \sin^3 \theta$ $= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta)$ $= \cos 3\theta + i \sin 3\theta$ Equating imaginary parts, $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$ $= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$ $= 3 \sin \theta - 4 \sin^3 \theta$	Award 2 correct solution Award 1 attempt to expand or to use De Moivre's Theorem.
(ii) E2	Let $x = \sin \theta$ and $\sin 3\theta = -\frac{1}{2}$. $-\frac{1}{2} = 3x - 4x^3$ $\therefore 8x^3 - 6x - 1 = 0$ Solutions of this equation are the solutions of $\sin 3\theta = -\frac{1}{2}$ $\therefore 3\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}$ $\therefore \theta = \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{19\pi}{18}$ $\therefore x = \sin \frac{7\pi}{18}, \sin \frac{11\pi}{18}, \sin \frac{19\pi}{18}$ $= \sin \frac{7\pi}{18}, \sin \frac{11\pi}{18}, \sin \left(-\frac{\pi}{18}\right)$ $= \sin \frac{7\pi}{18}, \sin \frac{11\pi}{18}, -\sin \frac{\pi}{18}$	Award 2 correct solution Award 1 establishes appropriate relationships
(iii) E4	$\therefore \sin \frac{7\pi}{18} + \sin \frac{11\pi}{18} + \left(-\sin \frac{\pi}{18}\right) = -\frac{b}{a} = 0$ $\therefore \sin \frac{7\pi}{18} + \sin \frac{11\pi}{18} = \sin \frac{\pi}{18}$	Award 1 correct solution

(b) (i) E4	$\therefore \alpha + (-\alpha) + \beta = -q$ $\therefore \beta = -q$ $\therefore P(-q) = (-q)^3 + q(-q)^2 + r(-q) + s$ $= -q^3 + q^3 - qr + s$ $= 0$ $\therefore s - qr = 0$ $\therefore s = qr$	Award 3 correct solution Award 2 find an expression for two of q, r and s in terms of α and β. Award 1 find an expression for one of q, r and s in terms of α and β.
(ii) E2 E4	Since $s = qr$ $P(x) = x^3 + qx^2 + rx + qr$ $= (x + q)(x^2 + r)$ So the roots of $P(x)$ are $-q$ and the roots of $x^2 + r = 0$. If $r = -\gamma$ then the roots are $-q$ and $\pm\sqrt{\gamma}$ $\Rightarrow$ all real $\therefore$ contradicts initial assumption. If $r = 0$ then the roots are $-q$ and 0 $\Rightarrow$ all real $\therefore$ contradicts initial assumption. Hence, $r > 0$ and two of the roots of $P(x)$ are the purely imaginary numbers $\pm i\sqrt{r}$.	Award 2 correct solution Award 1 observe that two of the roots are complex conjugates
(c) (i) E2 E3	$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n$ $= \left(1 + 2\cos^2 \frac{\pi}{n} - 1 + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}\right)^n$ $= \left(2\cos^2 \frac{\pi}{n} + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}\right)^n$ $= \left(2\cos \frac{\pi}{n} \left(\cos \frac{\pi}{n} + i \sin \frac{\pi}{n}\right)\right)^n$ $= 2^n \cos^n \frac{\pi}{n} (\cos \pi + i \sin \pi)$ $= -2^n \cos^n \frac{\pi}{n}$	Award 2 correct solution Award 1 attempts to use given trigonometric relationships
(ii) E2 E3	$(1 + i\sqrt{3})^{2n} + (1 - i\sqrt{3})^{2n}$ $= \left(2\text{cis} \frac{\pi}{3}\right)^{2n} + \left(2\text{cis} \left(-\frac{\pi}{3}\right)\right)^{2n}$ $= 2^{2n} \text{cis} \frac{2n\pi}{3} + 2^{2n} \text{cis} \left(-\frac{2n\pi}{3}\right)$ $= 2^{2n} \left(\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} + \cos \frac{2n\pi}{3} - i \sin \frac{2n\pi}{3}\right)$ $= 2^{2n} \times 2 \cos \frac{2n\pi}{3}$	Award 3 correct solution Award 2 substantial progress towards solution Award 1 limited progress towards solution

If n is a multiple of 3, $n = 3k$ where $k \in \mathbb{Z}$

$$\begin{aligned}\therefore 2^{2n+1} \cos \frac{2n\pi}{3} &= 2^{2n+1} \cos \frac{6k\pi}{3} \\ &= 2^{2n+1} \cos 2k\pi \\ &= 2^{2n+1} \cdot 1 \\ &= 2^{2n+1}\end{aligned}$$

If n is not a multiple of 3, $n = 3k - 1$ or $n = 3k + 1$ where $k \in \mathbb{Z}$

$$\begin{aligned}\therefore 2^{2n+1} \cos \frac{2n\pi}{3} &= 2^{2n+1} \cos \frac{2(3k-1)\pi}{3} \\ &= 2^{2n+1} \cos \left(2k\pi - \frac{2\pi}{3} \right) \\ &= 2^{2n+1} \cdot -\frac{1}{2} \\ &= -2^{2n}\end{aligned}$$

or

$$\begin{aligned}\therefore 2^{2n+1} \cos \frac{2n\pi}{3} &= 2^{2n+1} \cos \frac{2(3k+1)\pi}{3} \\ &= 2^{2n+1} \cos \left(2k\pi + \frac{2\pi}{3} \right) \\ &= 2^{2n+1} \cdot -\frac{1}{2} \\ &= -2^{2n}\end{aligned}$$